MedTrack_DV

Hospital Operations & Patient Analytics Dashboard

FINAL PROJECT DOCUMENTATION

Project Information	Details
Internship	**Infosys Springboard Virtual Internship 7.0**
Domain	**Data Visualization**
Project	**MedTrack_DV – Hospital Operations & Patient Analytics Dashboard**
Dashboard Platform	**Tableau Desktop**
Data Processing	**Python, Pandas, NumPy**
Prepared By	**Isha Kumari**
Mentor	**Shanmukha Priya**
Project Period	**June 2026 – August 2026**
Final Dashboard Suite	**Hospital Overview | Patient Flow | Department Analytics | Resource Utilization**

Module 8 – Documentation and Project Delivery

1. Project Overview

MedTrack_DV is a Tableau-based hospital operations and patient analytics project developed during the Infosys Springboard Virtual Internship 7.0. The project transforms healthcare and hospital operational data into an integrated set of interactive dashboards for monitoring admissions, patient flow, department performance and resource utilization.

1.1 Problem Statement

Hospitals generate large volumes of patient and operational information. Without structured analysis, it can be difficult to monitor admissions, patient movement, department performance, bed utilization and resource availability. MedTrack_DV addresses this through a centralized interactive Tableau dashboard suite.

1.2 Objectives

- Collect and integrate healthcare, patient, admission, department and resource datasets.
- Clean and standardize data using Python.
- Engineer hospital and department-level KPIs.
- Build four interactive Tableau dashboards.
- Integrate dashboards using navigation, filters and dashboard actions.
- Validate KPI calculations and dashboard functionality before final delivery.
- Prepare an organized, documented and portfolio-ready project.

2. Project Scope and Module 8 Requirements

The project brief defines Module 8 as Documentation and Project Delivery. The final documentation covers dataset sources, KPI definitions, dashboard guidance, healthcare operations methodology, project organization and final delivery.

Module	Focus	Output
1	Data Collection	Hospital/patient datasets and data collection script
2	Data Cleaning & Transformation	Cleaned dataset and cleaning script
3	Hospital KPI Engineering	KPI-ready final dataset and KPI generation script
4	Dashboard Planning & Prototyping	Storyboard and Tableau prototype
5	Hospital Overview & Patient Flow	Two developed Tableau dashboards
6	Department Analytics & Resource Utilization	Two developed dashboards and integrated workbook
7	Testing & Validation	QA Checklist and Dashboard Testing Report
8	Documentation & Project Delivery	Final Documentation, GitHub repository and Tableau workbook

3. End-to-End Project Workflow

The project follows the complete analytics workflow defined in the internship project brief:

DATA COLLECTION → DATA INTEGRATION → CLEANING & TRANSFORMATION → KPI ENGINEERING → TABLEAU DASHBOARDS → INTEGRATION → QA VALIDATION → FINAL DELIVERY

Raw hospital and patient datasets were collected and processed using Python. The cleaned and integrated data was used for KPI engineering and Tableau dashboard development. The final workbook was then tested and organized for GitHub delivery.

4. Data Sources and Dataset Structure

The final working project uses multiple hospital and patient-related datasets to support patient-flow, department and resource analysis.

Dataset / File	Purpose
patient.csv	Patient-level information
admission.csv	Admission and patient-flow analysis
department.csv	Department identification and comparison
doctor.csv	Doctor-related operational information
employee.csv	Staff and allocation analysis
bed.csv	Bed status, occupancy and availability analysis
disease.csv	Disease/category analysis
patient_diagnostic.csv	Patient diagnostic information
patient_insurance.csv	Patient-insurance linkage
insurance_provider.csv	Insurance provider analysis
department_kpis.csv	Department KPI and efficiency analysis
hospital_final_dataset.xlsx	Tableau-ready integrated dataset
hospital_raw_data.csv	Raw/integrated project data output
hospital_cleaned.csv	Cleaned project data output

5. Data Collection and Integration

Data collection and dataset loading were performed using Python and Pandas. The project contains data_collection.py for the collection/loading stage.

1. Load available healthcare and hospital CSV datasets using Pandas.
2. Inspect row counts, columns and data types.
3. Check identifiers required to connect patient, admission, department and resource data.

4. Prepare datasets for cleaning and integration.
5. Export integrated outputs for KPI engineering and Tableau.

5.1 Integrated Dataset

The project integrated multiple operational datasets into a Tableau-ready analytical structure. The final integrated dataset used for dashboard development should be documented below using the exact final file/version used in the submitted workbook.

Final Dataset	Records	Fields	File
Tableau-ready integrated dataset	85724	53	hospital_final_dataset.xlsx

6. Data Cleaning and Transformation

Data cleaning was implemented in Python. The project brief requires duplicate removal, missing-value handling, department standardization, healthcare-indicator normalization and Tableau-ready outputs.

Cleaning Step	Purpose
Duplicate handling	Remove duplicate records to prevent inflated counts.
Missing-value handling	Identify and handle missing patient/operational fields.
Date conversion	Convert date fields into usable dates for time-based analysis.
Text standardization	Remove extra spaces and standardize categorical values.
Department standardization	Keep department categories consistent.
KPI preparation	Create fields required for hospital, patient-flow and department analysis.
Output generation	Export cleaned/integrated data for Tableau.

7. KPI Engineering and Definitions

The project brief requires six major KPI areas: Total Admissions, Occupancy Rate, Average Length of Stay, Readmission Rate, Bed Utilization Rate and Department Efficiency Score.

KPI	Calculation / Logic	Validated Value
Total Admissions	Count of unique Admission IDs	45,000
Average Length of Stay	Average Length of Stay across relevant admissions	4.144
Occupancy Rate	Occupied Beds ÷ Total Beds × 100	69.88%
Bed Utilization Rate	Utilized Beds ÷ Available Bed Capacity × 100	69.88%
Readmission Rate	Readmitted Patients ÷ Eligible Patients × 100	56.92%
Department Efficiency Score	Use the exact formula/logic implemented in generate_hospital_kpis.py	Department-level

7.1 Department Efficiency Score Results

Department	Efficiency Score
Emergency	3560.26
ICU	1663.90
Internal Medicine	3127.51
Orthopedics	2419.08
Pediatrics	3424.79
Surgery	4100.14

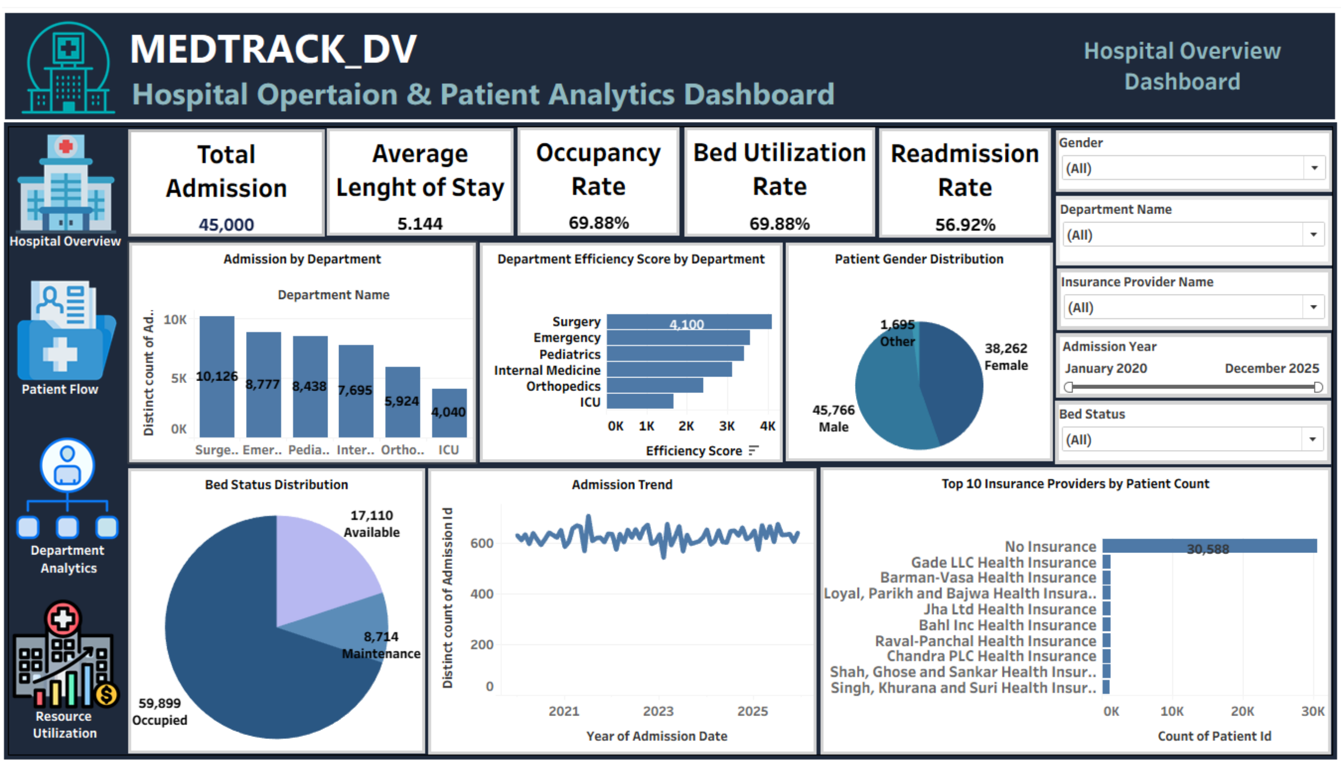
8. Dashboard Development Guide

The final Tableau workbook contains four interconnected dashboards designed around the requirements in the internship project brief.

8.1 Dashboard 1 – Hospital Overview

Purpose: Provide a high-level summary of hospital performance and operational metrics.

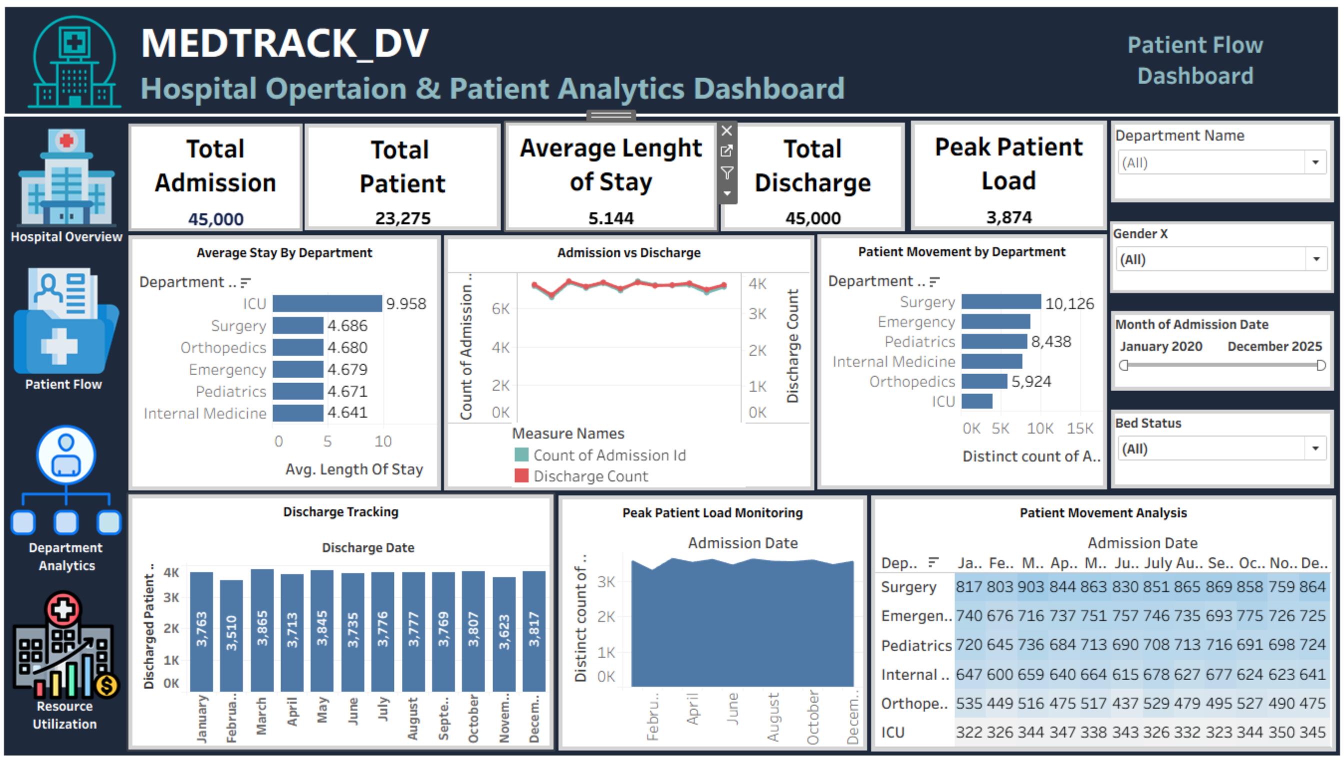
- Hospital KPI cards.
- Admissions and operational trends.
- Occupancy and bed utilization monitoring.
- Readmission analysis.
- Department comparison and filtering.
- Interactive dashboard controls.



8.2 Dashboard 2 – Patient Flow

Purpose: Track patient movement through admissions, discharges and departments and identify high-load periods.

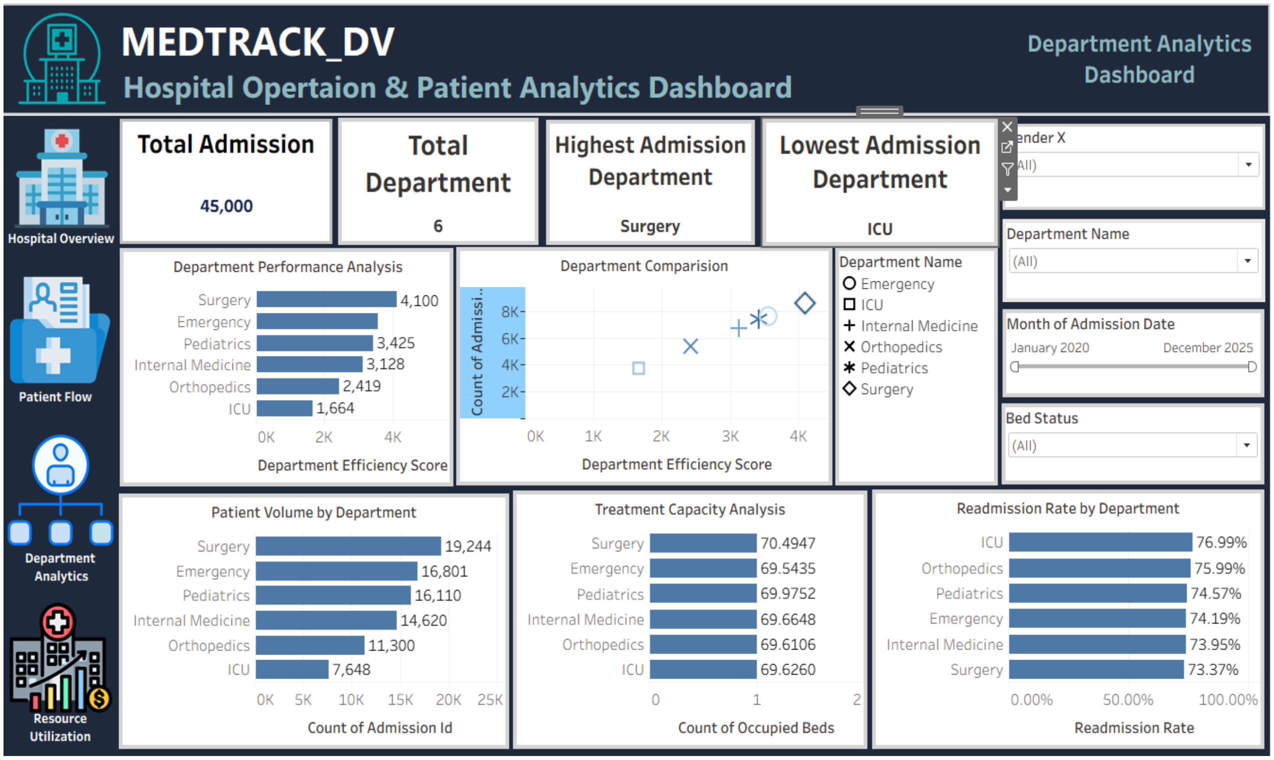
- Admission trends.
- Discharge tracking.
- Admission versus discharge comparison.
- Patient movement analysis.
- Average stay analysis.
- Peak patient-load monitoring.
- Interactive time/department filters.



8.3 Dashboard 3 – Department Analytics

Purpose: Compare departmental performance and identify high-volume, high-performing and improvement areas.

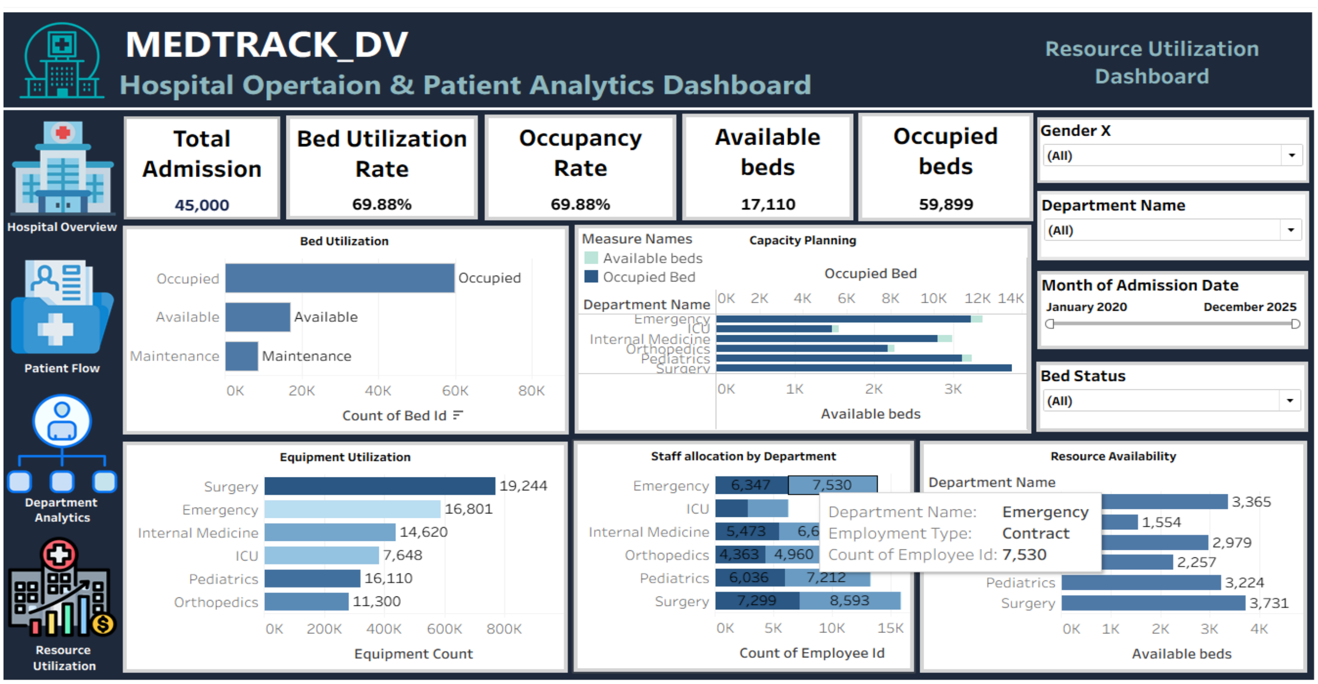
- Best-performing department.
- Highest-admissions department.
- Lowest-readmission department.
- Patient volume by department.
- Department efficiency comparison.
- Treatment capacity analysis.
- Department comparison scatter plot.



8.4 Dashboard 4 – Resource Utilization

Purpose: Monitor hospital resource utilization and capacity planning.

- Bed utilization analysis.
- Staff allocation monitoring.
- Resource/equipment analysis where supported by the dataset.
- Capacity planning insights.
- Resource availability analysis.
- Department-level resource comparison.



9. Dashboard Integration and User Guide

The four dashboards were integrated into a single Tableau workbook. Navigation controls, filters, tooltips and dashboard actions support interactive exploration.

Feature	User Action	Purpose
Dashboard Navigation	Select a navigation button.	Move between dashboards.
Home Navigation	Use Home button.	Return to Hospital Overview.
Department Filter	Select a department.	Focus analysis on the selected department.
Date/Month Filter	Select a period.	Analyze operational trends.
Tooltips	Hover over a mark.	View detailed values.
Dashboard Actions	Select interactive chart elements.	Explore related views.

10. Healthcare Operations Methodology

The dashboard suite is organized around four operational areas:

Operational Area	Question Supported	Dashboard
Hospital Performance	What is the overall hospital operational status?	Hospital Overview
Patient Flow	How are admissions, discharges and patient movement changing?	Patient Flow
Department Performance	Which departments have higher volume, efficiency or readmission concerns?	Department Analytics
Resource Utilization	How effectively are beds and resources being utilized?	Resource Utilization

The dashboards support descriptive and operational decision-making. They are intended for analytics and monitoring, not for clinical diagnosis or treatment decisions.

11. Testing and Validation Summary

Module 7 testing was completed before final delivery. The final documentation should match the exact test counts in the submitted QA Checklist and Dashboard Testing Report.

Testing Area	Cases	Passed	Failed
KPI Validation	6	6	0
Hospital Overview	6	6	0
Patient Flow	5	5	0
Department Analytics	5	5	0
Resource Utilization	4	4	0
Integration & Final Workbook	5	5	0
Overall	31	31	0

11.1 KPI Validation Status

- Total Admissions – PASS
- Average Length of Stay – PASS
- Occupancy Rate – PASS
- Readmission Rate – PASS
- Bed Utilization Rate – PASS
- Department Efficiency Score – PASS

12. Development Issues and Resolutions

Issue	Resolution
Admission aggregation initially reflected integrated rows instead of unique admissions.	Corrected Tableau aggregation to use the appropriate Admission ID level.
Small KPI calculation differences during development.	Validated final Tableau calculations against Python results.
Department filter initially affected unintended worksheets.	Configured filter scope to the intended worksheets.
Some dashboard visualizations required adjustment.	Corrected worksheet configuration, aggregation and dashboard actions.

13. Final Project Structure

```
MedTrack_DV/
├── Data/
├── Scripts/
├── Dashboard/
├── Documents/
└── LICENSE
```

Folder	Contents
Data	CSV/Excel datasets, cleaned data, integrated data and KPI data.
Scripts	data_collection.py, hospital_cleaning.py, generate_hospital_kpis.py.
Dashboard	Final Tableau workbook and dashboard-related Tableau files.
Documents	Final documentation, QA checklist, testing report and project presentation.

14. Final Deliverables

- Final Tableau workbook (.twbx) containing the four integrated dashboards.
- hospital_final_dataset.xlsx – Tableau-ready integrated dataset.
- department_kpis.csv – department KPI dataset.
- data_collection.py – data collection/loading script.
- hospital_cleaning.py – cleaning and transformation script.
- generate_hospital_kpis.py – KPI generation script.
- QA Checklist – Module 7.
- Dashboard Testing Report – Module 7.
- Final Project Documentation – Module 8.
- Project Presentation.
- GitHub repository with organized project folders.

15. Key Project Outcomes and Insights

- The project created a centralized view of hospital operations and patient activity.
- The KPI layer supports monitoring of admissions, length of stay, occupancy, utilization, readmission and department efficiency.
- The four dashboards provide complementary views of hospital, patient, department and resource operations.
- Dashboard navigation, filters, tooltips and actions were tested successfully.
- Python and Tableau results were cross validated during QA.

Insight 1: The hospital recorded 45,000 total admissions, reflecting high patient activity.

Insight 2: Bed occupancy and utilization remained stable at 69.88%, indicating efficient resource management.

Insight 3: The Surgery department achieved the highest efficiency score (4100.14), while the readmission rate was 56.92%, highlighting opportunities for patient-care improvement.

16. Limitations and Future Scope

The current solution is based on the available datasets and represents an analytical dashboard rather than a live hospital information system.

- Connect dashboards to real-time hospital data sources.
- Add automated refresh pipelines.
- Add patient-demand and admission forecasting.
- Develop predictive readmission models.
- Introduce advanced resource optimization and capacity forecasting.
- Add more granular staff and equipment utilization when reliable operational data is available.

17. Conclusion

MedTrack_DV demonstrates an end-to-end healthcare analytics workflow from data collection and cleaning to KPI engineering, interactive Tableau dashboard development, integration, testing and final documentation. The final dashboard suite provides a structured view of hospital operations, patient flow, department performance and resource utilization. The completed QA and validation activities support the reliability of the final workbook for internship demonstration and portfolio use.

18. Final Documentation Sign-off

Field	Details
Prepared By	Isha Kumari
Project	MedTrack_DV
Module	Module 8 – Documentation and Project Delivery
Documentation Status	Completed
Dashboard Suite	Four integrated Tableau dashboards
QA Status	PASS